



Chapter 17 – Confusion

Episode overview:

- 1) List the Major Categories for the differential diagnosis of Confusion
- 2) Differentiate between organic and functional confusion.
- 3) List the Critical and Emergent causes of Confusion

Wise Cracks:

- 1) Describe a quick assessment tool for screening for confusion

Rosen's in Perspective:

Confusion means many different things, but we are particularly interested in processes impacting:

- Memory
- Attention
- Awareness

Remember, confusion is a *symptom*, not a diagnosis – it reflects *recent change in behaviour*. Do not overlook subtle confusion in the elderly, as it is an independent risk factor for increased mortality. Confusion is a spectrum, ranging from:

- Mild impairment of short-term memory
- Delirium - a global inability to relate to environment and process sensory input:

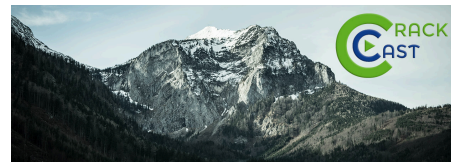
Hyperactive Delirium	Hypoactive Delirium
Acute confusional state with triad of: ▪ Increased alertness ▪ Increased psychomotor activity ▪ Disorientation +/- hallucinations	Acute confusional state with: ▪ Decreased alertness ▪ Decreased behaviour

Varied Terminology for Confusion:

- Altered mentation
- Change in mental status
- Altered mental status
- Change from baseline
- "Different"

Pathophysiology

Again as discussed in chapter 16, Consciousness is made up of **arousal** and **content**, which are controlled by the ARAS and cerebral cortex respectively. With confusion, we are really talking about issues with *content*, or *cortical disruption*.



Three major categories can impact and disrupt normal cortical function:

- I. **Substrate deficiency**
 - a. Hypoxia
 - b. Hypoglycemia
- II. **Neurotransmitter dysfunction**
 - a. Endocrine disease
 - b. Hepatic failure
 - c. CNS sedatives
 - d. EtOH
 - e. Poisons
- III. **Circulatory dysfunction**
 - a. Shock

1) List the Major Categories for the differential diagnosis of Confusion:

Again, we can go back to the “DIMS” (DIMES see episode 16) approach or use the categories that Rosen’s uses:

Box 17-1 Major Categories: Differential Considerations

Primary intracranial disease
 Systemic diseases secondarily affecting the central nervous system
 Exogenous toxins
 Drug withdrawal

I like to use the DIMS approach - it’s simple and I can use this memory aid for MANY presenting complaints (seizures, AMS, confusion, syncope, etc).

- The key is to use BROAD categories

2) Differentiate between organic and functional confusion.

A more classical Rosen’s question would probably ask you to contrast organic vs. functional confusion (psychiatric) by listing 6-10 features.

Break it down like Rosen’s does based on:

1. History
 - a. Sudden onset (hrs to days) vs. gradual
 - b. Age - vs. middle aged
2. Mental status exam
 - a. Fluctuating LOC vs. alertness
 - b. Disorientation vs. oriented
 - c. Inattention vs. agitation/anxiety
 - d. Visual hallucinations vs. auditory
 - e. Cognitive difficulties vs. delusions/illusions

Box 17-2 Findings That May Help Differentiate between Organic and Functional Causes of Confusion

Organic	Functional (Psychiatric)
History	History
Acute onset	Onset over weeks to months
Any age	Onset ages 12 to 40 years
Mental status examination	Mental status examination
Fluctuating level of consciousness	Alert
Disoriented	Oriented
Attention disturbances	Agitated, anxious
Poor recent memory	Poor immediate memory
Hallucinations: visual, tactile, auditory	Hallucinations: most commonly auditory
Cognitive changes	Delusions, illusions
Physical examination	Physical examination
Abnormal vital signs	Normal vital signs
Nystagmus	No nystagmus
Focal neurologic signs	Purposeful movement
Signs of trauma	No signs of trauma



3. Physical exam

- a. ABNORMAL VITAL SIGNS vs. normal VS
- b. Nystagmus vs. NO nystagmus
- c. Focal neurological signs vs. purposeful behaviour
- d. Signs of trauma vs. no signs of trauma

A little pearl here for you: KNOW this list well.

When you're seeing any patient and especially one triaged to the psychiatric Emergency Area you should be looking for and documenting these pertinent positive and negative findings in your exam section!

3) List the Critical and Emergent causes of Confusion

Box 17-3 Critical and Emergent Diagnoses

Critical

Hypoxia, diffuse cerebral ischemia
Respiratory failure
Congestive heart failure
Myocardial infarction
Shock
Systemic processes
Hypoglycemia
CNS infections
Hypertensive encephalopathy
Elevated intracranial pressure—medical and surgical origin

Emergent

Hypoxia, diffuse cerebral ischemia
Severe anemia
Systemic diseases
Electrolyte and fluid disturbance
Endocrine disease
Thyroid
Adrenal
Hepatic failure
Nutrition, Wernicke's encephalopathy
Sepsis, infection
Intoxications and withdrawal
CNS sedatives
Ethanol
Other medication side effects, particularly anticholinergics
CNS disease
Trauma
Infections
Stroke
Subarachnoid hemorrhage
Epilepsy, seizures
Postictal state
Nonconvulsive status epilepticus
Complex partial status epilepticus
Neoplasm

Note: These represent a partial diagnosis; causes are myriad. "Critical" in this case means conditions that need immediate assessment and correction within moments, such as oxygenation and ventilation problems or hypoglycemia. Because confusion represents CNS failure, other problems may be considered critical as well and may necessitate intensive care unit admission, depending on severity.

CNS, central nervous system.

Critical:

Shock and hypoxia
Hypoglycemia
Head infections
Hypertensive encephalopathy
Head exploding

Emergent:

Anemia leading to diffuse cerebral ischemia

Metabolic diseases

- Electrolytes
- Endocrine
- Hepatic encephalopathy/failure
- Nutrition
- Sepsis
- Drugs and poisons
- Intracranial
 - Trauma
 - Infection
 - Stroke
 - SAH
 - Tumour
 - Seizures

- Don't forget non-convulsive status epileptics or epileptic twilight state)



Remember the 4 H's or the phrase "the Hyper Hippo's Head"

- Hyper (hypertensive encephalopathy)
- Hypoglycemia
- Hypoxia
- Hypotension
- Head
 - Tumour, Stroke, SAH, Seizure, etc.

All that being said there is a lot of overlap with this in the last chapter - if there's truly something big bad and ugly going on - the patient will exhibit altered mental status, rather than just confusion.

Wise Cracks

So you have the confused patient: you have collected a history with GOOD collateral information, now it's time for your head to toe exam.

1) Describe a quick assessment tool for screening for confusion

ITEM	SCORE (highest number in category indicates correct response; decreased scoring indicates increased number of errors)	WEIGHT	SCORE
What year is it now?	0 or 1 (score 1 if correct; 0 if incorrect)	x2	
What month is it?	0 or 1 (score 1 if correct; 0 if incorrect)	x2	
Repeat phrase and remember it: "John Brown, 42 Market Street, New York"			
About what time is it? (answer correct if within the hour)	0 or 1 (score 1 if correct; 0 if incorrect)	x2	
Count backward from 20 to 1	0, 1, or 2 (score 2 if correct; 1 if 1 error; score 0 if more than 2 errors)	x1	
Say the months in reverse	0, 1, or 2 (score 2 if correct; 1 if 1 error; score 0 if more than 2 errors)	x1	
Repeat the memory phrase (each underlined portion is worth 1 point)	0, 1, 2, 3, 4, 5 (score 5 if correctly performed; each error drops score by one)	x1	
		TOTAL	_____

Final score is sum of the totals; score less than 15 suggests the presence of altered cognition and need for further assessment.

Figure 17-2. Quick Confusion Scale.

Don't forget: almost everyone with confusion should be getting the veterinary workup:

- Full set of vitals
- Glucose
- CBC, lytes, liver panel, kidney function
- ECG (MI)
- Urine
- CXR
- CT head (maybe) and LP (maybe)